

Practical 5 - Collapsibility

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

Datasets required: `simdata_binary.dta`, `simdata_binary2.dta`

Introduction

This practical is in three parts. In Part A we return to the kidney stones example used in session 4. Parts B and C use simulated data.

Aims

1. To explore relationships between marginal and conditional treatment effect estimates, considering both collapsible and non-collapsible quantities.
2. To gain further practice at interpreting marginal and conditional treatment effect estimates.
3. To extend the scenarios considered in the notes to include a continuous treatment-outcome confounder.

Part A: Kidney stones example

In session 4 and practical 4 we estimated the marginal and conditional treatment effects (of X on Y) using risk differences, risk ratios and odds ratios. See Tables 4.2 and 4.3.

Question 1

In the notes we saw that the marginal risk difference is equal to a weighted average of the conditional risk differences (conditional on stone size Z). What are the weights? Show that the marginal risk difference is equal to a weighted average of the conditional risk differences.

Question 2

In the notes we saw that the marginal risk ratio is equal to a weighted average of the conditional risk ratios (Conditional on stone size Z). What are the weights? Show that the marginal risk ratio is equal to a weighted average of the conditional risk ratios.

Question 3

Can the marginal odds ratio be expressed as a weighted average of the conditional odds ratios? Explain your answer.

Discuss: What are the interpretations of the marginal and conditional risk differences, risk ratios and odds ratios.

Discuss: For a patient with a large kidney stone, which treatment effect estimate(s) are most relevant to inform their treatment?

Part B: A simulated example with binary outcome

In this part we some simulated data (`simdata_binary.dta`) on a binary treatment X , binary outcome Y and binary covariate Z . The covariate Z is known to be measured temporally prior to X , meaning that we can be sure it does not lie on the causal pathway between X and Y .

Question 4

Load the data using `haven::read_dta` (or with `use simdata_binary.data` in Stata). Then quantify the relationship between X and Z .

Question 5

Using a logistic regression that includes X , Z and their interaction, estimate the conditional probabilities $\Pr(Y = 1|X = x, Z = z)$ for $x = 0, 1$ and $z = 0, 1$, and hence the conditional risk differences

$$\Pr(Y = 1|X = 1, Z = z) - \Pr(Y = 1|X = 0, Z = z), \quad z = 0, 1$$

Question 6

Without performing any further calculations, what do you expect the marginal risk difference to be?

Question 7

From your logistic regression in question 5, what are the estimates of the conditional odds ratios

$$\frac{\Pr(Y = 1|do(X = 1), Z = z)/\Pr(Y = 0|do(X = 1), Z = z)}{\Pr(Y = 1|do(X = 0), Z = z)/\Pr(Y = 0|do(X = 0), Z = z)}, \quad z = 0, 1$$

Question 8

Fit another logistic regression to obtain an estimate of the marginal odds ratio

$$\frac{\Pr(Y = 1|do(X = 1))/\Pr(Y = 0|do(X = 1))}{\Pr(Y = 1|do(X = 0))/\Pr(Y = 0|do(X = 0))}$$

What assumption do you make? Explain the difference between this estimate and your estimates in question 7.

Part C: A simulated example with binary outcome and continuous confounder

In this part we use some simulated data (`simdata_binary2.dta`) on a binary treatment X , binary outcome Y and continuous variable Z , where Z confounds the association between X and Y .

Question 9

Fit a logistic regression of Y on X and Z and their interaction. What is the conditional odds ratio for an individual with (a) Z equal to its median value in the data, (b) Z equal to its 10th percentile in the data, (c) Z equal to its 90th percentile in the data.

Question 10

Using the empirical standardization method introduced in session 4, manually obtain an estimate of the effect of X on Y using a marginal odds ratio. Then use the `avg_comparisons` function in the `marginaleffects` package (or `margins` in Stata) to check your result.

Question 11

Compare this with the odds ratio from a regression of Y on X alone. What is the reason for the difference between this estimate and that in question 10?

Question 12

Discuss: What are the interpretations of the marginal and conditional odds ratios?